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Deep neural network based approach for ECG classification using hybrid differential features and active learning

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Kazım Hanbay[1] ✉

[1]*Bingol University, Computer Engineering Department, Bingol 12000, Turkey*
✉ *E-mail: khanbay@bingol.edu.tr*

Abstract: A novel active learning-based electrocardiogram (ECG) signal classification method using eigenvalues and deep learning is proposed. Six statistical features relating to ECG beat intervals are calculated separately for each heartbeat. Both statistical features and eigenvalues of ECG beats are combined into a single feature vector. The eigenvalues of ECG beats are used as an input to denoising autoencoder (DAE). Weighted ECG beat intervals are calculated by using ten-fold cross-validation approach. To learn an efficient feature representation from the hybrid feature vector, DAE is used in an unsupervised way. After completing the feature learning procedure, a softmax regression layer is added on the top of the resulting hidden layer of DAE, and thus a suitable deep neural network (DNN) architecture is built. The learned features obtained from the autoencoder layers are fed to the softmax regression layer for classification. To update weights of the proposed eigenvalues-based DNN model, ECG beats are labelled by the medical expert are used. In order to determine the most informative beats, entropy and Breaking-Ties are also used as selection criteria. The proposed method is evaluated in terms of ECG beats classes. The classification performance of the authors' proposed model is also compared with the several conventional machine learning classifiers.

1 Introduction

Electrocardiogram (ECG) signals reveal the electrical activities of the cardiac system using some surface electrodes. These signals include different amplitudes and frequencies when heart disease occurs [1]. The ECG signal analysis has an important role in disease diagnosis such as cardiac arrhythmias. To reflect the underlying heart conditions, the behaviours of the ECG signal should be carefully analysed. Cardiologists can realise signal pattern differences but in some cases, two ECG signals have similar electrical activity patterns but indicate different cardiac diseases. Therefore only using the raw ECG signals waveform is not an adequate treatment to determine the cardiac diseases. The following questions should be addressed at this point: How is the information hidden in the ECG signals? Can these signals be analysed in more detail? What are the different components they contain? A more accurate diagnosis will be made when these questions are answered.

In general, the ECG signals contain six different frequency components. These components are called P, Q, R, S, T and U. These components need to be carefully analysed for a qualified diagnosis and treatment. To diagnose correctly, specific features of these signals should be extracted by using pattern recognition approaches. Thus computer-based approaches ensure a reliable solution for cardiologists.

According to the Association for the Advancement of Medical Instrumentation (AAMI), the ECG signals define five kinds of cardiac diseases: unknown beats (Q), normal (N), ventricular (V), supraventricular (S) and fusion of normal and ventricular (F). The important point here is that the ECG signal is analysed and one of the five diseases is detected correctly. At this point, machine learning-based methods are used to find out all the distinctive features in the signal.

In the last decades, various ECG signal feature extraction methods have been developed in the literature such as kernel models [2], geometric features [3, 4], morphology and heartbeat interval features [5, 6], temporal information [3], principal component analysis [7, 8] and hidden Markov modelling [9]. These models are not acceptable since these methods do not consider the non-linear behaviour of the ECG signal information. Thus, they are insufficient for ECG classification.

Discrete wavelet transform (DWT) is another method to analyse the ECG signal frequencies [10]. The DWT provides a time-scale representation of the ECG signals, which is very important for non-stationary signals, such as the ECG signal. Li *et al.* [11] presented a dyadic wavelet transform for obtaining ECG characteristic points. The proposed model first detects the QRS complex, then the T wave, and finally the P wave. To find sharp change points in the ECG signal, many wavelets at different scales are used.

Alternatively, other methods, such as the discrete Fourier, cosine transform and statistical features have been used to calculate features from the ECG signals in both the frequency and the time domains [12–14]. Then, the obtained features are fed as input for classification algorithms. Among these models, artificial neural network (ANN) and support vector machines (SVMs) are commonly used with the hand-crafted features [15–17].

In some cases, automatic methods do not make good classification in terms of recommendations of the AAMI. Thus AAMI recommends inter-patient scenario for performance evaluation of methods. In this scenario, it is necessary to record the ECG signals from different patients for testing and training. This strategy is not usually considered in most of the literature works. Thus, the automatic classification of ECG signals very challenging due to the strong shift and small amplitude differences (mV) between the distributions of test and training signals of different patients.

To overcome the above issues, expert-knowledge-based (i.e. semi-automatic) methods have been developed as alternative solution [18, 19]. In these methods, a global classifier is trained using a large dataset. Then, a local classifier is trained using the ECG test signal labelled by a medical expert. Voting rules are used and, a customised classifier is designed for each patient using local and global classifiers. Since patient-specific ECG signal data is classified, the most specific and effective treatment can be applied [20].

It should be noted that the ECG signals may be corrupted by large amounts of noise from electrode motion and muscle artefacts. To extract useful features from the ECG signals, it is important to apply appropriate signal denoising algorithms. Several adaptive recurrent filters such as a normalised least mean square and signed regressor algorithm can be used for ECG signal denoising [21].

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Recently, deep neural networks (DNNs) have also been applied successfully in multi-dimensional signal processing problems. The deep learning provides a good feature representation from the input data [22]. Various deep learning architectures include stacked autoencoder (SAE) [23], convolutional neural networks (CNNs) [24] and deep belief networks (DBNs) [25]. Compared to traditional learning and classification models, DNNs have achieved acceptable results in time-series data [26, 27], object recognition [28] and medical image analysis [29]. Especially, time-series classification approaches have to deal with high-dimensional data. Moreover, it is difficult to calculate all distinctive and proper features for obtaining representative properties which encode the input data.

Therefore, various deep learning based techniques have been proposed to extract key features using time-series data. In general, two different approaches have been adopted for multi-dimensional signal classification applications [30]. The first approach uses DNNs as a feature extractor that is combined with generative/discriminative approaches, while the second approach uses DNNs as an entire classification framework. It should be noted that the first approach is more successful in terms of the analysing of QRS complexes in the ECG signals. On the other hand, the second approach cannot separate similar ECG signals. Since ECG signals have dynamical nature, these signals cannot be classified using a fixed framework DNN.

This paper suggests an unsupervised approach for active classification of ECG signals using a DNN model created from an improved denoising autoencoder (DAE) reformed by eigenvalues of ECG signals. The basic idea of this method is to provide the higher level feature representation to lower hidden layers by the differential features of ECG signals. These interpretable features significantly reduce the classification error while a patient-specific diagnosis model is developed.

In the context of the proposed ECG signal analysis, DAE and eigenvalues are used to achieve four significant goals: (i) extract an appropriate feature of the ECG signals in terms of P, Q, R, S, T components instead of using hand-crafted features; (ii) benefit from the active learning (AL) methodology to reduce the medical expert requirement in labelling ECG signal beats for training the classifier; (iii) the proposed eigenvalues-based DNN feature learning approach is integrated with other conventional classifiers, namely the ANN, Naive Bayes (NB), linear SVM, random forest (RF) and decision tree (DT); and (iv) the feature learning procedure is developed to be capable of generating relevant features without requiring a feature selection method, as opposed to traditional feature extraction methods. Using the available training data, a DAE is trained with the eigenfeatures of ECG signals to learn an optimal feature representation and find the optimal network weights in an unsupervised way. More precisely, the eigen-based ECG signal features are used as input to DAE instead of raw ECG signal data. After training the DAE, the new ECG signals are sent through the eigenvalue analysis, and then into the trained DAE to classify the ECG signals. The output of DAE is ended with a softmax regression layer to classify ECG signals according to the AAMI classes. During the interaction stage, it should be noted that the proposed method does not perform the user label procedure in the first few minutes. AL methods are used to rank the unlabelled ECG data.

2 Related works

2.1 Eigenvalues of ECG signal data

Eigenvalues and associated eigenvectors are necessity computations for many engineering problems such as signal processing, texture classification and computer vision [31–33]. They can be used to calculate the directional rate of change of signal [33]. From the point of view of the ECG classification problem, eigenvalues have a great significance on the sensitive detection of T wave and QRS wave groups. However, affected by electromagnetic noises and interference, ECG signals contain different noises, causing the location problems of ECG signal components. The accurate approach is to first compute the eigenfeatures of T wave and QRS wave groups and find the significant frequency components during feature extraction. Several methodologies have been developed to remove noise by using eigenvalues corresponding to noisy components [34].

Table 1 Generation of separable two-dimensional filters from the one-dimensional filters

Basic filters	Filter in $\boldsymbol{x}$	Filter in $\boldsymbol{y}$
$\boldsymbol{G_x}$	f_1	f_2
$\boldsymbol{G_y}$	f_2	f_1
$\boldsymbol{G_{xx}}$	f_3	f_2
$\boldsymbol{G_{xy}}$	f_4	f_4
$\boldsymbol{G_{yy}}$	f_2	f_3

Eigenvalues of the ECG signal can be calculated using a Gaussian first derivative filter and the Gaussian second derivative filter. According to [33, 35], these filters could be rotated at any angle by a linear combination of two basis filters and three basis filters. Suppose that the Gaussian function is defined as

$$G(x, y) = e^{\left(-\left(x^2+y^2\right)/\sigma^2\right)} \tag{1}$$

The four basic one-dimensional Gauss filters are given as [33]

$$\begin{aligned} f_1 &= \frac{-2t}{\sigma^2} e^{-(t^2/\sigma^2)} \\ f_2 &= e^{-(t^2/\sigma^2)} \\ f_3 &= \frac{2t}{\sigma^2}\left(\frac{2t^2}{\sigma^2} - 1\right) e^{-(t^2/\sigma^2)} \\ f_4 &= \frac{2t}{\sigma^2} e^{-(t^2/\sigma^2)} \end{aligned} \tag{2}$$

Table 1 denotes the generation step of filter responses. In this paper, three basic filters (i.e. G_{xx}, G_{xy} and G_{yy}) are only used to calculate eigenvalues of the ECG signal.

Let us consider $\boldsymbol{X} = (x_1, x_2, x_3, \ldots, x_n)$ an ECG signal vector of dimension D. The eigenvalues of the ECG signal are calculated as [33]

$$\begin{aligned} \lambda_{1,2} = &\pm\sqrt{\frac{\left(\boldsymbol{X} * G_{xx} - \boldsymbol{X} * G_{yy}\right)^2}{4} + \left(\boldsymbol{X} * G_{xy}\right)^2} \\ &+ \frac{\boldsymbol{X} * G_{xx} + \boldsymbol{X} * G_{yy}}{2} \end{aligned} \tag{3}$$

where $*$ denotes the convolution, G_{xx}, G_{yy} and G_{xy} are the second partial derivatives of $G(x, y)$ with regard to x and y. λ_1 and λ_2 are the eigenvalues of the ECG signal.

The concept of eigenvalues can help us to understand the role of different frequency components in ECG signals. Eigenvalues are the minimum and maximum of the normal curvature at a given point on a geometric surface [36]. Since eigenvalues are intrinsic and important geometric properties, they give a more descriptive information in the detection of T wave and QRS wave groups of ECG signals.

2.2 Denoising autoencoder

Motivated by building an effective ECG signal representation, deep learning approaches are used to learn the high-level frequency information from ECG signals. Thanks to a large number of hidden layer nodes, DAE provides both sparse representation and dimensionally reduction. DAE is used as an unsupervised feature learning mechanism, including the input, hidden and the output layers after decoding. To achieve noisy robustness, the original ECG signal $\boldsymbol{x}_i \in \Re^D$ is corrupted by adding Gaussian noise.

In general, DAE is composed of encoding and decoding parts, respectively. In the encoding layer, the noisy input $\tilde{\boldsymbol{x}}_i$ is mapped to the hidden layer representation $h_i \in \Re^L$ with the non-linear

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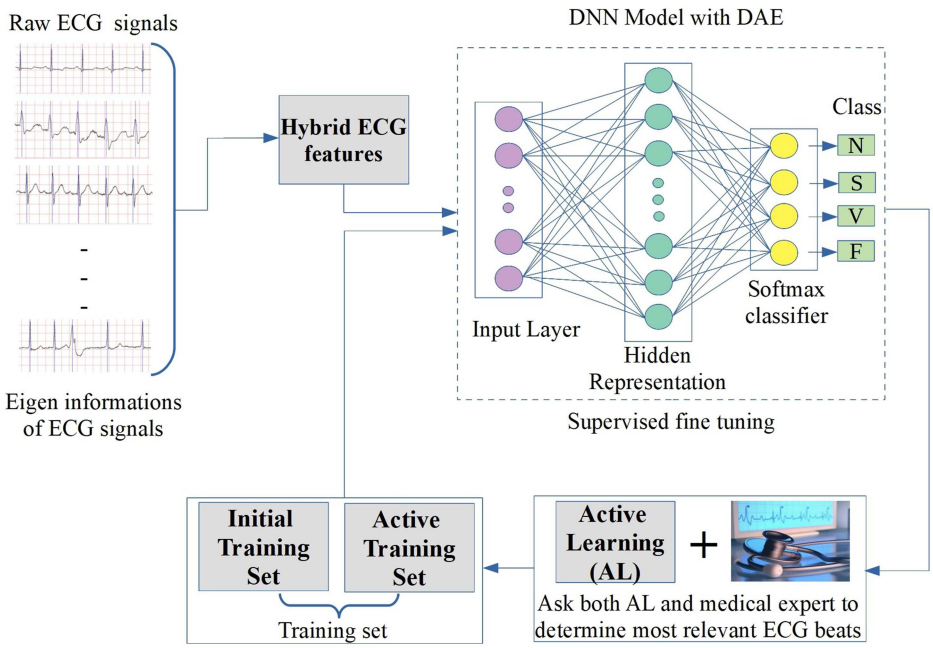

Fig. 1 *Flowchart of the proposed method*

activation function f. Hidden layer representation $h_i = f\left(\boldsymbol{W}^{(e)} \cdot \tilde{\boldsymbol{x}}_i + \boldsymbol{b}^{(e)}\right)$ is computed from the vector $\tilde{\boldsymbol{x}}_i$, where $\boldsymbol{W}^{(e)} \in \Re^{L \times D}$ is the encoder matrix. $\boldsymbol{b}^{(e)} \in \Re^{L}$ is the encoder bias vector. $f(.)$ denotes the activation function. In this paper, the sigmoid function is selected as the activation function. Similar to the encoding layer representation, the decoding layer performs the hidden layer representation through a nonlinear sigmoid activation function $r_i = f\left(\boldsymbol{W}^{(d)} \cdot h_i + \boldsymbol{b}^{(d)}\right)$. $\boldsymbol{W}^{(d)} \in \Re^{D \times L}$ and $\boldsymbol{b}^{(d)} \in \Re^{D}$ denote the decoder weight matrix and the decoder bias vector, respectively.

The parameter vector of DAE network is calculated by minimising the following cost function [27]:

$$\mathscr{L}_1(\boldsymbol{\theta}_{\mathrm{DAE}}) = \frac{1}{2n}\sum_{j=1}^{n} \| \boldsymbol{x}_i - r_i \|_2^2 + \frac{a_1}{2}\left(\| \boldsymbol{W}^{(e)} \|_F^2 + \| \boldsymbol{W}^{(d)} \|_F^2\right) + a_2 \sum_{j=1}^{L} \mathrm{KL}(\rho \parallel \hat{\rho}) \tag{4}$$

Here, $\boldsymbol{\theta}_{\mathrm{DAE}}$ is the parameter vector. $\left(\boldsymbol{W}^{(e)}, \boldsymbol{W}^{(d)}\right)$ denote the weight variables for encoder and decoder, respectively. a_1 and a_2 are constant regularisation parameters. $\mathrm{KL}(\rho \parallel \hat{\rho})$ is the Kullback–Leibler (KL) divergence. KL divergence can be calculated between a sparsity target ρ and average activation $\hat{\rho}_j$ as an additional penalty term

$$\sum_{j=1}^{L} \mathrm{KL}(\rho \parallel \hat{\rho}) = -\sum_{j=1}^{L}\left[\rho \log\frac{\rho}{\hat{\rho}_j} + (1-\rho)\log\frac{1-\rho}{1-\hat{\rho}_j}\right] \tag{5}$$

Equation (4) includes three significant terms. The first term denotes the construction error on the input training examples. The second and third terms are the weight penalty and the sparsity constraint terms, respectively. These terms are used to avoid the over-fitting problem. Also, they provide sparse representations and generalisation ability.

To optimise the parameters of the DAE model in (4), the parameter vector $\boldsymbol{\theta}_{\mathrm{DAE}}$ is initialised to small values. Then, the L-BFGS method [37] is applied to minimise the average reconstruction error. It should be noted that DAE architecture with the hidden layer is trained through the greedy layer-wise unsupervised way. This approach yields a hierarchical features learning mechanism. In generally, (4) is optimised by using the original input data. In this way, the first DAE is trained in an unsupervised mode, and the first hidden representation layer is obtained. To train the next DAE for higher-level representations, the reconstruction layer in the first DAE is removed. Thus hidden representation layer in the second DAE is obtained and employed as the ECG input data [27]. The learned features are presented as input to a linear classifier. Linear classifiers need to retrain procedure for the new input samples. For ECG classification problems, this approach is not efficient, and it is a time-consuming process. Thus, an alternative eigenvalues-based approach is developed in the next phases.

3 Methodology

In this paper, a new ECG classification method – inspired by the deep learning paradigm – is developed that combines eigenvalues and DAE. An illustration of the improved deep learning model, which contains the DNN architecture and the eigenfeatures of ECG signal is depicted in Fig. 1. We develop a general methodology taking into account some 'a priori' information on the ECG components that provides important advantages in the classification task. The main approach is a modification of deep learning classification methodology by considering eigenvalues-based hybrid feature extraction framework. The eigenvalues-based ECG feature representations are fed as input to DAE. The proposed approach comprises of three main phases: (i) unsupervised feature extraction and learning using eigenvalues and DAE; (ii) supervised fine tuning of DNN model; and (iii) DNN fine tuning with AL.

3.1 Unsupervised feature extraction and learning

This study proposes a novel method for classifying ECG signals using a DNN created from an improved DAE reformed by eigenvalues, taking advantage of the describing local structures of ECG signals. In this paper, DAE is firstly used for learning the features of a dataset in an unsupervised manner. It needs to be emphasised that the proposed method uses the eigenvalue magnitude of ECG signal instead of the raw ECG signal as inputs to DAE. This approach is the most innovative aspect of our method.

A training set $S = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}$ consists of n ECG beats, where $\boldsymbol{x}_i \in \Re^{D}$ is the ECG beat vector in dimension D, and $y_i \in [1, 2, 3, \ldots, C]$ denotes the corresponding class label. Firstly, the eigenvalues of the ECG signal $\boldsymbol{x}_i$ are calculated by using (3). The magnitude of the eigenvalues is calculated as follows:

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$$M(\lambda_1, \lambda_2) = \sqrt{\lambda_1^2 + \lambda_2^2} \quad (6)$$

where λ_1 and λ_2 are the eigenvalues of ECG signal. The DAE is trained to approximate and reconstruct the eigenvalue magnitude signal of the original ECG signal. $M(\lambda_1, \lambda_2)$ is given as input to a DAE.

In addition to the magnitude of the eigenvalues, weighted interval features (e.g. RR, S-T and P-R) are adopted to learn the underlying structure of the training ECG data.

An ECG signal has a P wave, QRS-complex and T wave. The start and end points of P, Q, R, S, and T waves provide considerable convenience in diagnosing the disease. These significant features have a larger signal variation rate than other signal regions [4]. To provide an extensive signal analysis of the ECG beats through DAE, six statistical features of the ECG waveform are firstly calculated [27]. These features are added to the training set since they reveal the class signatures. To define sequential ECG beats, $R(t)$ and $R(t-1)$ denote the current R peak and previous R peak, respectively. Also, $R(t+1)$ denotes the next R peak. The first statistical feature is the Pre-RR information. Pre-RR is calculated using the distance between $R(t)$ and $R(t-1)$. The second statistical feature is the Post-RR information. Post-RR is calculated using the distance between $R(t)$ and $R(t+1)$. The third statistical feature is the local RR information. Local RR is calculated by averaging all the RR intervals in the past 10 s part of the current ECG record. The fourth statistical feature is the global RR information. It is calculated using the average information of RR intervals. The last two statistical features are the QRS-start and QRS-end points. These intervals are determined using a sliding window covering the past 5 min part of the current ECG record [27, 38].

To provide robust signal recognition, both weighted interval features and raw ECG interval features are used in a hybrid manner. In this paper, weighted PR, RR and ST intervals are also calculated to enhance the patient-specific classification performance. It should be pointed that PR, RR and ST interval features do not directly use as in [27, 39]. For revealing the impact of the weights of PR, RR and ST intervals on classification performance and finding the optimal weight values of PR, RR and ST intervals, ten-fold cross-validation approach is used based on the MIT-BIH database. The weights values ranging from 1 to 45 are tested. Figs. 2*a–c* illustrate the classification accuracies of three weighted intervals in terms of weight values. From Fig. 2, we observe that the weights ranging from 9 to 17 have higher classification success. It can be concluded that any of 14 or 15 is the most appropriate value to weight these intervals. After the appropriate value is determined, three weighted interval features are combined with the previously calculated six statistical properties. As a result, a single statistical feature vector is built. The length of the initial feature vector equals to 9 for each ECG beat at this stage.

In addition to the statistical features, the morphology of complete ECG waveform is also extracted. Thus, wave boundary recognition and QRS detection tasks of ECG signals are performed. The heartbeat segments are obtained by using the annotations of beat locations presented in the MIT-BIH database. Segmented ECG signals are uniformly resampled as similar signal samples. In this way, morphological features are obtained. Periodic lengths of these samples equal to 50. The length of the initial feature vector $\boldsymbol{s}$ fed as input to DAE including 9 statistical and 50 morphological features equals to 59 for each ECG beat. Fig. 3*a* shows the important components of the ECG signal such as P, Q, R, S, T waves and QRS-start, QRS-end points. Also, Fig. 3*b* shows the Pre-RR, Post-RR, P-R and S-T intervals.

To construct a discriminant feature vector, both eigenvalue magnitude $M(\lambda_1, \lambda_2)$ and raw ECG interval information are calculated at the same time. Then, both magnitude and interval features are combined into a hybrid single vector $\boldsymbol{v} = (M(\lambda_1, \lambda_2), s)$. To learn the features in an unsupervised manner, the proposed eigenvalues-based feature extraction mechanism is integrated with DAE. The calculation steps of the hybrid feature vector given as input to the DAE are shown in Fig. 4. The input vector $\boldsymbol{v}$ is mapped

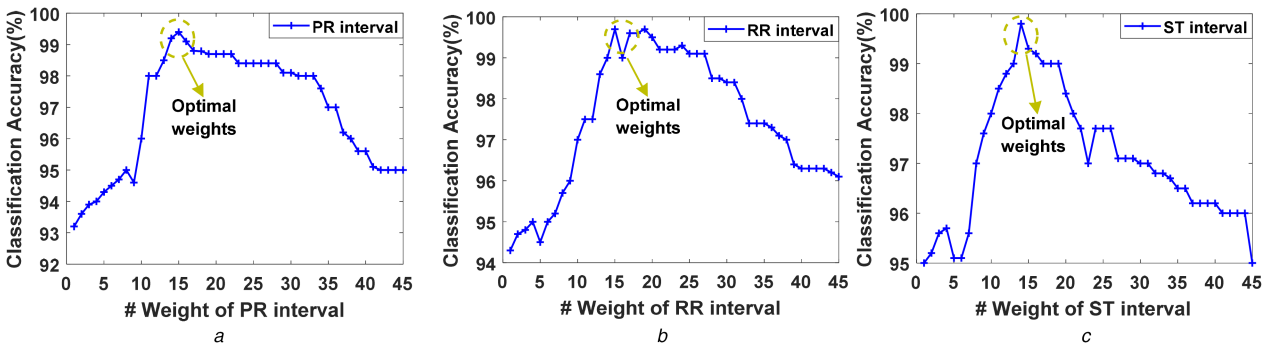

Fig. 2 *Determination of the significant ECG component points;*
(a)–(c) Ten-fold cross-validation for selecting the weight of the PR, RR and ST intervals

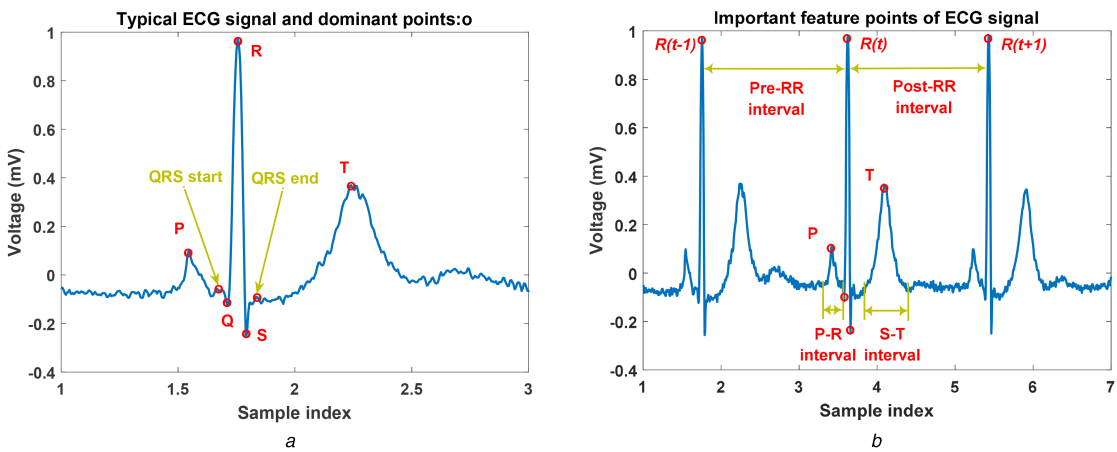

Fig. 3 *Typical ECG signals*
(a) ECG waveform of one cardiac cycle and the principal components of the ECG signal, ***(b)*** Important components of the ECG signal

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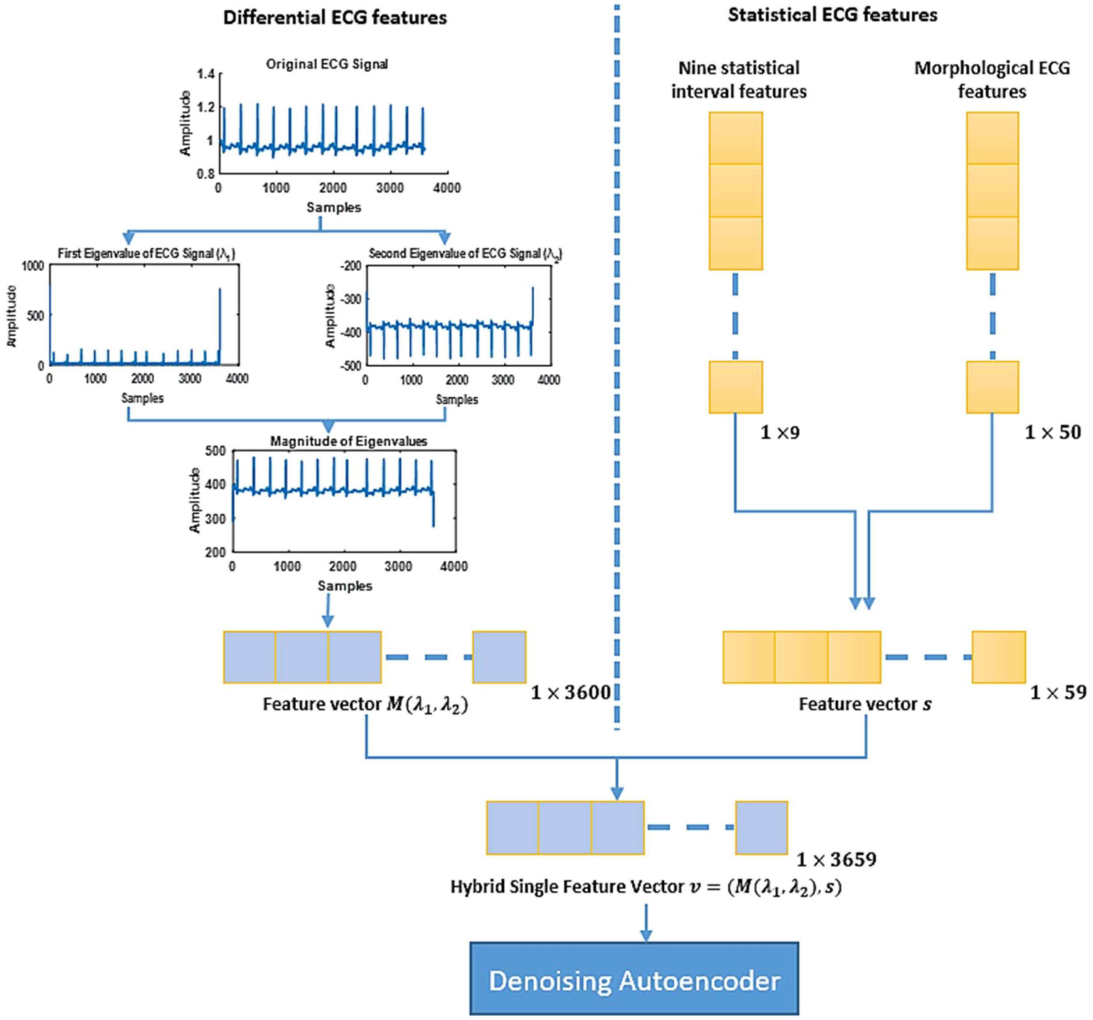

Fig. 4 *Calculation steps of the proposed hybrid features given as input to the DAE*

to $x \in [0, 1]^p$ as the input of the DAE by min–max normalisation. The DNN is created with two hidden layers using the DAE for ECG signal classifying. DAE is trained to minimise the loss function and find the optimal parameters. The obtained feature vector is fed into the classifier.

3.2 Supervised fine tuning of DNN model

In the previous section, pretraining step based on greedy layer wise is performed. To build a task-specific supervised learning schema, a softmax classifier layer is added on the final hidden representation layers of DAE. Softmax layer carries out efficient classification procedure for both binary and multiclass problems. ECG classification applications deal with four class classification problem according to the AAMI standard. Thus, a fine-tuned DNN architecture is optimised using the backpropagation algorithm. The following cost function is used to minimising network error [27]:

$$\mathscr{L}_2(\theta_{\mathrm{DNN}}) = -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} 1(y_i = k)\log\left(\frac{\exp(h_{\theta_{\mathrm{DNN}}}(x_i))}{\sum_{k=1}^{K}\exp(h_{\theta_{\mathrm{DNN}}}(x_i))}\right) + \frac{a_3}{2n}\left(\| W_{\mathrm{softmax}} \|_F^2 + \sum_{l=1}^{H} \| W_l \|_F^2\right) \tag{7}$$

Here the first term defines cross-entropy loss in the softmax layer. The weight decay penalty term is defined in the second term. $h_{\theta_{\mathrm{DNN}}}(x_i)$ denotes the output of the DNN model for an input $\boldsymbol{x}_i$, and $1(.)$ is an indicator function.

In this stage, all parameters of DNN model (i.e. $\theta_{\mathrm{DNN}} = W_1, \ldots, W_H, b_1, \ldots, b_H, W_{\mathrm{softmax}}$) are estimated by initialising the weights of the softmax regression layer while the hidden layer weights are initialised by the encoding weights calculated in the pretraining stage. Finally, the cost function in (7) is minimised using a gradient descent algorithm. As a result, the improved DNN finds the optimal network parameters.

3.3 DNN fine tuning with AL

To avoid potential misdiagnosis, ECG classification works need to evaluate heart beats by a medical expert since ECG signals are commonly used to diagnose tool for determining cardiac disasters. Labelling by the medical expert increases the classification accuracy. Generally, better classification performance is obtained if the pre-trained DNN is fine-tuned rather than only used as a feature extractor. Fine-tuning strategies are also used to obtain some discriminative information of the dataset. Using an iterative procedure, this paper uses a robust approach based on AL which benefits from a medical expert to find the most important ECG test beats.

Basically, AL is used to train a classifier as fast as possible with few labels. What is more, AL can automatically determine the most relevant data to query their labels. Thus, the cost of training (i.e. labelling) time needed to learn a given cost function is considerably reduced. AL algorithms use two types of query selection criteria, i.e. representativeness and informativeness [40]. The data determined by these criteria preserves the data distribution.

AL approach determines a proper criterion for ranking all ECG beats [27]. The ranking is performed according to the contribution of the ECG beats in the classification task. Then, the most informative and relevant ECG beats are selected. These beats are reserved for adding to the active set. These relevant ECG beats have an effective role in fine tuning the DNN classifier.

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To select the most informative ECG beats, state-of-the-art AL approaches, including entropy score [27, 40] and Breaking-Ties (BT) are considered. To build either an effective classifier, only a small subset of the ECG signals is used. This is performed by selecting those ECG signals from the unlabelled ECG signals that are affirmed to be most informative for labelling by a medical expert [41]. In entropy criterion, the entropy score of each ECG test beat is calculated. The E_{AL} beats with the highest entropy scores are chosen for labelling. ECG beats with the highest entropy scores are classified with low confidence, and thus they are added to the training set for improving the classifier decision regions [27]. Thus, classifier performance is improved since a reliable classifier is built. Another AL approach BT is used to improve the confidence of the multi-class classification [42]. In BT criterion, the training sample selection bases on the posterior probabilities. BT selects the ECG samples with either the smallest classification confidence or the smallest difference between the two highest probability classes [43]. The E_{AL} beats with the lowest difference between the two highest posterior values are obtained through BT for learning.

The most important disadvantage of AL-based classifiers is that they easily overfit, whereas they converge quickly with less training samples. To train an accurate DNN prediction model, previous and current iterations of the DNN are fused using the max rule [27]. By applying this approach, an efficient DNN model is obtained for classifying ECG signals. The main steps of the proposed model called eigenvalues-based active-DNN are described in Algorithm 1.

4 Experimental results

This paper investigates the ECG classification problem under three experiments: (i) eigenvalues-based DNN model is applied to extract the relevant features from the raw ECG signals and to classify them into the four classes; (ii) only the first two layers of the DNN network (i.e. DAE architecture with hidden layer) are used to calculate the feature vectors and then an ANN is used to reveal the difference between ANN and DNN classifiers in ECG classification, (iii) similar to experiment 2, only the first two layers of the DNN network are used to calculate the feature vectors and then we used other conventional machine learning classifiers, namely NB, linear SVM, RF and DT. In all the experiments in this paper, it must be noted that six statistical features relating to ECG beat intervals are calculated separately for each heartbeat. Both statistical features and eigenvalues of ECG beats are combined into a single feature vector. The eigenvalues of ECG beats are used as an input to stacked DAE.

In order to evaluate the performance of the eigenvalue-based deep features, ECG records are collected from the MIT-BIH arrhythmia database [44]. This database consists of 48 records from 47 different patients. Each ECG beat has two-lead signals and is ~30 min in length. In this study, the 40 records are selected to evaluate the proposed model. The first 20 records are used as representative beats to be included in the common training data and the remaining 24 records are selected as a test set. Similar to [27, 38], the four recordings (102, 104,107, 217) with paced beats are excluded from the test and training sets. All models are trained and tested on a single NVIDIA Titan Z GPU (6 GB RAM) and an Intel i7 4770 CPU (@ 3.40 GHz).

ECG signals should be classified into five classes according to AAMI standards: N, V, S, F and Q. However, class Q is not considered since it is marginally represented in the databases. The main purpose is to distinguish the VEB (V class versus [N, S and F]) and SVEB (S class versus [N, V and F]) from others.

Algorithm 1: Eigenvalues-based active-DNN

Inputs:
- Training ECG set $S = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}$.
- Test ECG record: $R = \{(x_j)\}_{j=1}^{m}$.
- Initialised DNN parameters
- AL iterations: *ITER*
- Number of ECG beats for labelling at each iteration: E_{AL}

Output:
- Final ECG classification result

Step 1: Compute the eigenvalues of ECG beats by using (3);
Step 2: Compute the magnitude of eigenvalues by using (6), $M(\lambda_1, \lambda_2)$;
Step 3: Compute nine statistical interval features of ECG beat, $s_{1\times 9}$;
Step 4: Compute the morphological features of ECG beat, $s_{1\times 50}$;
Step 5: Combine the differential and statistical interval features to reconstruct hybrid feature vector, $\boldsymbol{v} = (M(\lambda_1, \lambda_2), s)$;
Step 6: Use a hybrid feature vector as input data for learning of DAE in an unsupervised manner;
Step 7: Compute the weights θ_{DAE} of DAE by optimising the cost function in (4) using L-BFGS method; *(unsupervised feature extraction and learning)*
Step 8: Use θ_{DAE} to initialise DNN architecture and calculate θ_{DNN} by optimising the cost function in (7) using backpropagation algorithm *(supervised fine tuning)*
Step 9: Classify the test ECG record R with the trained DNN; let PP denotes the estimated posterior probabilities;
Step 10: Constitute the initial active training set to an empty set: $T_{AL} = \varnothing$;
Step 11: *for iter* $= 1 : ITER$ *(DNN fine tuning with AL)*

Step 11.1: Compute the entropy and BT criteria of each test ECG beat using the estimated PP;
Step 11.2: Rank the ECG signals of the test record R according to their entropy and BT uncertainty values;
Step 11.3: Label the top ranked signals by asking both AL and medical expert;
Step 11.4: Update the T_{AL} by using these labelled signals; New $T_{AL} = [\, T_{AL}; Selected_{T_{AL}}]$;
Step 11.5: Update the weights of DNN by fine tuning on T_{AL};
Step 11.6: Classify the test record R by using the updated DNN; $PPnew$ corresponds to the current estimated posterior probabilities;
Step 11.7: Update the DNN posterior probabilities by means of the max rule: $PPupdated = max(PP, PPnew)$, $PP = PPupdated$;
Step 11.8: Set the labels of the test record R through the maximum of PP;
end
Step 12: Final ECG classification result.

Similar to literature works, the pre-processing operation is first applied using a 200 ms width median filter. To remove high-frequency noisy and the baseline wander, a 12-order low-pass filter is applied. Then the initial training set is built typical ECG records from the following 22 ECG signals grouped into DS1 = {101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230} [38]. To obtain a patient-specific DNN architecture, various representative beats (i.e. QRS complexes) are selected as patient specific data from each ECG signal. The remaining 22 ECG records of MIT-BIH are used as test data. Test data is grouped into DS2 = {100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234}.

Four standard metrics are used to evaluate the classification performance of the DNN model: positive predictive value (Pp), sensitivity (Se), specificity (Sp) and overall accuracy (OA).

4.1 Initial DNN model

In this paper, a fine-tuned DNN model that includes the DAE and the softmax regression layer is used to classify the ECG signals. DAE includes the input, hidden and the output layers after decoding. The valuable recommendations in [27, 45] are considered to train DNN model. The number of hidden layers and the number of hidden neurons are selected through a series of error and trial experiments. There is one hidden layer, and it contains 100 hidden neurons. The sigmoid function is used as the activation function.

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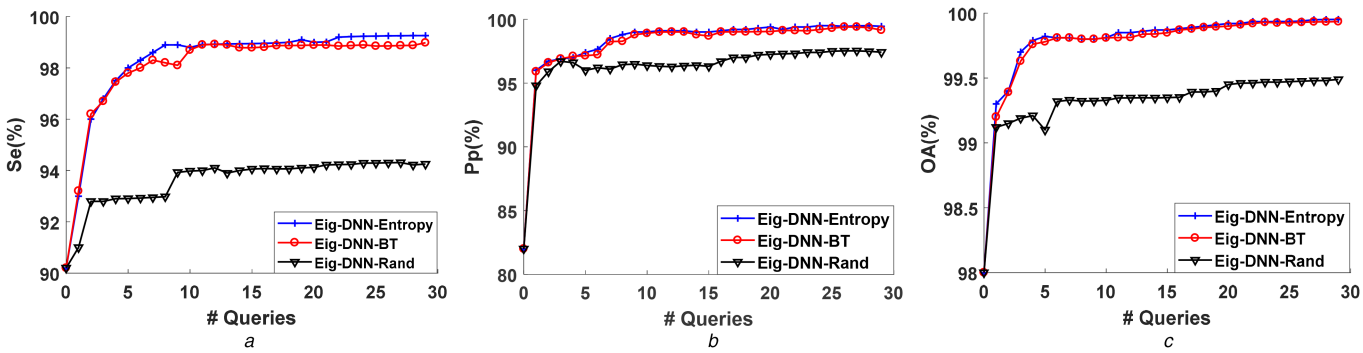

Fig. 5 *VEB classification performances of the proposed models for DS1 + DS2 of MIT-BIH*
(a) Se, ***(b)*** Pp and, ***(c)*** OA

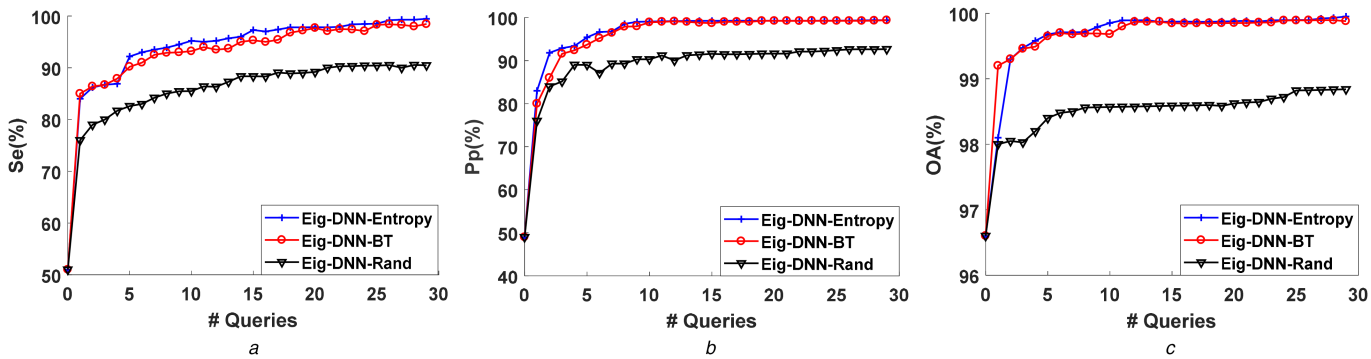

Fig. 6 *SVEB classification performances of the proposed models for DS1 + DS2 of MIT-BIH*
(a) Se, ***(b)*** Pp and, ***(c)*** OA

Firstly, DAE is pre-train on DS1 of MIT-BIH using L-BFGS optimisation algorithm. Secondly, pre-training results of DAE are employed to initialise the DNN. The weights of DNN architecture are updated with the backpropagation algorithm. The parameter θ_{DAE} of DAE is randomly initialised to small values in the range $[-0.0045\ \ 0.0045]$. The sparsity target activation function of the hidden unit is set to $\rho = 0.05$. To optimise the cost function in (4), the function *minFunc* is used with default parameters. The mini-batch gradient optimisation approach optimises the backpropagation algorithm. In the mini-batch gradient optimisation approach, batch size is set to 100, learning rate to 1 and momentum to 0.5. Then using a five-fold cross-validation method, a searching process is applied to find the hyperparameters of DNN architecture (i.e. λ_1, λ_2 and λ_3).

4.2 Experiment I: eigen-based DNN

In this section, the proposed eigen-based DNN model is applied to ECG signal classification. Both testing and training signals are taken from the MIT-BIH database [44]. All experiments are done according to AAMI standards. Three different scenarios are considered to deal with the AAMI ECG classification. Three evaluation datasets are constructed. For VEB and SVEB detection, evaluation dataset 1 contains 11 testing records (200, 202, 210, 213, 214, 219, 221, 228, 231, 233, 234) and 14 testing records (200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234), respectively. Evaluation dataset 2 contains 24 testing records (200 and onwards). Evaluation dataset 3 contains all of 44 records (i.e. DS1 + DS2).

To emphasise the benefits of using entropy score and BT metrics, ECG beats obtained by random selection from dataset as done similar works are classified [27]. The classification results are compared with the results of the ECG signals selected by the entropy score and BT criteria. In the rest of the paper, the proposed deep learning models are called Eig-DNN-Entropy, Eig-DNN-BT and Eig-DNN-Rand.

In Figs. 5 and 6, the plots of accuracy parameters are shown in terms of the number of queries for DS1 + DS2. It should be pointed out that the medical expert labels ten ECG beats from the test record for each query. To provide a stable learning, labelled ECG beats are inserted into the AL training set. In this way, the eigen-based DNN is trained by the backpropagation optimisation algorithm, and the network weights are updated. As shown in Figs. 5 and 6, Eig-DNN-Entropy and Eig-DNN-BT achieve the best results for all performance parameters. For the VEB classification, the initial values of Se, Sp, Pp and OA are equal to 86.3, 97.1, 89.2 and 96.8%. After ten iterations (100 beats), Eig-DNN-Entropy and Eig-DNN-BT achieve classification accuracies of (93.4, 99.5, 94.6 and 98.72%) and (94.7, 98.21, 97.35 and 98.6%). Eig-DNN-Rand obtains the worst accuracies of (93.2, 93.58, 96.38 and 78.6%). From the distribution of the accuracies according to the number of queries, it is clear that ECG beats ranked by entropy score and BT have shown best accuracy performing curve as compared to random selection method. Another important point to highlight is that the ranked beats are clinically important since they provide patient-specific treatment method.

For SVEB classification, the proposed methods with ranked beats achieve maximum classification accuracies. Eig-DNN-Rand obtains the accuracy of 97.2%. The Se, Sp and Pp of Eig-DNN-Rand are 86.41, 96.4 and 97.3%, respectively. According to Figs. 5 and 6, it can be noticed that if the beat ranking is done on the dataset, the classification of ECG signals is performed more accurately, especially for detecting QRS peaks. The proposed eigen-based DNN model fuses both the statistical features of the ECG waveform and eigenfeatures of the raw ECG signal. To detect abnormalities in ECG data, this multi-linear feature representation provides promising results. Compared with Eig-DNN-Rand and previous works, the proposed model presents an outstanding performance in both VEB and SVEB detection.

The classification performance of the proposed DNN model is further compared with four other DNN models for the three different datasets, as shown in Tables 2–4. The first method uses the temporal information of ECG signal morphology [6]. This approach offers a customised ECG beat classifier. The second method uses morphological wavelet transform features [19]. For the pattern recognition unit, a feed-forward and fully connected ANN model is employed. The third method uses recurrent neural networks and density-based clustering approach [38]. The last method uses deep learning model for active ECG classification

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Table 2 Classification results of the proposed method and comparison with the state-of-the-art methods

Method	Labelled	SVEB				VEB			
		Se	Pp	Sp	OA	Se	Pp	Sp	OA
de Chazal and Reilly [6]	300	86.8	49	89.3	96.2	93.4	95.2	96.5	98.2
Ince *et al.* [19]	300	82.3	65.7	97.6	95.7	95.3	90.1	97.5	96.8
Al Rahhal *et al.* [27]	300	95.1	99.1	100.0	99.7	97.5	99.5	99.8	99.7
Zhang *et al.* [38]	201	99.4	97.6	99.7	97.6	98.7	87.4	99.4	89.4
Eig-DNN-entropy	0	14.5	13.8	96.7	91.2	85.6	88.7	98.4	96.9
	100	87.6	92.4	98.5	98.7	96.8	98.7	99.3	99.8
	200	94.5	98.6	99.7	99.5	97.2	99.6	99.8	99.6
	300	94.8	99.3	100.0	99.8	99.8	99.7	100.0	99.8
Eig-DNN-BT	0	16.8	20.7	94.4	92.5	86.9	90.2	99.2	96.4
	100	94.3	97.9	98.7	99.6	98.4	99.5	99.9	99.9
	200	96.3	99.7	99.8	99.5	98.7	99.7	99.2	99.1
	300	98.3	98.8	99.3	99.6	99.5	99.5	99.4	99.1

Table 3 Classification results of the proposed method and comparison with the state-of-the-art methods

Method	Labelled	SVEB				VEB			
		Se	Pp	Sp	OA	Se	Pp	Sp	OA
Ince *et al.* [19]	300	62.7	58.4	97.5	95.3	85.3	83.6	97.5	97.1
Al Rahhal *et al.* [27]	300	92.4	93.1	99.7	99.4	97.2	98.8	99.8	99.6
Zhang *et al.* [38]	204	99.6	97.5	99.8	97.9	98.9	86.7	99.5	89.0
Eig-DNN-entropy	0	40.4	43.5	97.4	95.6	92.4	91.4	98.7	98.5
	100	92.5	96.2	99.5	99.4	98.2	99.6	99.8	99.5
	300	93.4	99.5	100.0	99.5	99.6	99.8	100.0	99.6
Eig-DNN-BT	0	42.6	45.3	97.6	95.7	91.6	89.8	98.4	98.1
	100	95.4	96.9	99.8	99.7	97.7	99.2	99.4	99.8
	300	98.9	99.4	99.5	99.4	99.4	98.9	98.9	99.7

Table 4 Classification results of the proposed method and comparison with the state-of-the-art methods

Method	Labelled	SVEB				VEB			
		Se	Pp	Sp	OA	Se	Pp	Sp	OA
Ince *et al.* [19]	300	63.2	53.4	98.4	97.6	84.5	86.8	97.6	98.2
Al Rahhal *et al.* [27]	300	95.9	98.9	100.0	99.9	99.3	99.7	99.9	99.9
Zhang *et al.* [38]	198	99.7	97.1	99.9	98.1	99.3	85.9	99.7	88.7
Eig-DNN-entropy	0	91.1	95.4	98.7	97.6	92.4	85.9	98.3	97.6
	100	95.8	96.8	99.8	99.7	98.6	99.8	100.0	99.9
	300	99.2	99.5	100.0	99.9	99.8	99.9	100.0	99.9
Eig-DNN-BT	0	43.0	44.6	97.8	96.8	92.1	85.6	99.1	98.2
	100	94.3	97.2	99.7	99.6	98.7	99.6	99.9	99.9
	300	96.2	98.6	99.1	99.2	99.2	99.5	99.8	99.8

[27]. It obtains discriminative features from the raw ECG data using DAE. To rank each test beat in the ECG, the method uses the entropy score and BT approaches.

Similar to [27], Se, Sp, Pp and OA parameters are calculated for each method. At first, several experiments are conducted on the MIT-BIH database to determine the optimal labelling number. Then, it is observed how the different labelling numbers affect the results of Se, Sp, Pp and OA parameters. Labelling numbers are set to 0, 50, 100, 200 and 300, respectively, and obtained different performances, which are shown in Tables 2–4. Clearly, this work achieves numerically more accurate classification results than the other four methods. If the labelling number of ECG beats is under 150, the other three methods are not suitable in complex ECG classification. Similar to the proposed methodology, Rahhal *et al.* [27] developed a DNN architecture based on AL and softmax regression layer for the ECG signal classification. Thus, the ECG beats having high ambiguous are given to the medical expert for labelling. Then, these beats are employed to retrain the softmax classifier. This AL methodology improves the generalisation ability of a classifier, and thus their method has better results compared to the other three previous methods. Our proposed method classifies the ECG beats with a relatively high accuracy. Since the proposed model uses eigen-based deep learning to extract the structural and statistical properties of QRS peaks, the AL of feature representation by this method is stable to the complicated factors such as electromagnetic noises and interference. In addition to eigen-based raw ECG features, the proposed method calculates the dynamic interval features of ECG signals.

4.3 Experiment II: ANN classifier

The first two layers of the DNN network (i.e, DAE architecture with input and hidden layer) are used to calculate the feature vectors. In experiment II, it should be pointed out that DAE has linear activation function and the hidden layer has 559 neurons. Then an artificial neural network (ANN) is used to reveal the difference between ANN and DNN classifiers. The obtained features from the hidden layer of the DAE are used to train ANN classifier.

The neural network (NN) used in this experiment is a multilayer feed-forward NN with 559 inputs. ANN architecture consists of three layers (input, hidden, and output layers). The number of hidden layers and the number of hidden neurons are determined based on a series of trial experiments [30]. The number of neurons

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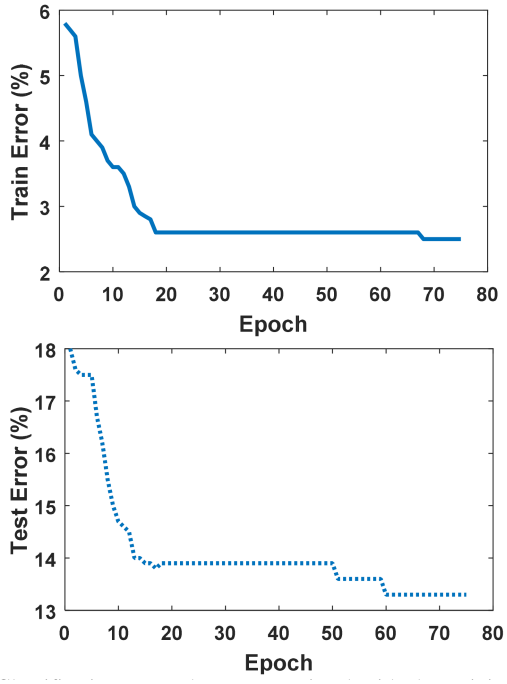

Fig. 7 *Classification error changes associated with the training and testing ECG data (i.e. DS1 + DS2 of MIT-BIH) by ANN classifier*

Table 5 Percentage of classification errors associated with the training and testing ECG data (i.e. DS1 + DS2 of MIT-BIH) by using the ANN classifier in terms of different number of neurons in the hidden layer

Number of hidden layer neurons	Training data % classification error	Testing data % classification error
7	2.95	12.42
12	2.84	12.21
17	2.85	11.45
22	2.88	11.28
27	2.71	11.55
32	2.85	12.42
37	2.84	11.21
55	2.90	12.42

in the hidden layer is obtained as 27 with the learning rate of 0.089. The output layer has three nodes, which shows the four types of ECG beats classes. The sigmoid function is used as the activation function. The training and the testing ECG data error classification results for the ANN classifier are shown in Fig. 7.

The minimum training error is achieved at Epoch = 61, and thus the learning process is terminated. To analyse the classification performance of the ANN classifier, Table 5 presents the relationship between the number of hidden layer neurons and classification error. As shown in Table 5, the minimum training data classification error is obtained by using 27 neurons in the hidden layer.

In addition to comparisons based on four accuracy parameters, a class-based assessment is also performed between the standard NN and the DNN. In the class-based assessments, the DS1 dataset is used. The classification results are shown in Fig. 8. It shows that the proposed eigenvalues-based DNN achieves more accurate results when the obtained hybrid features (PR, RR and ST intervals and eigenvalue-based features) are given as input to the DNN model. The overall accuracy for N, S, V, F classes of the proposed DNN model is 99.51%. Compared to standard feed-forward NN, there is a notable difference in the overall accuracy for four classes. The worst class accuracy is obtained for the V beat (96.15%). The most important point to mention is that class-based assessment is not reliable to determine patient-specific treatment procedure. For

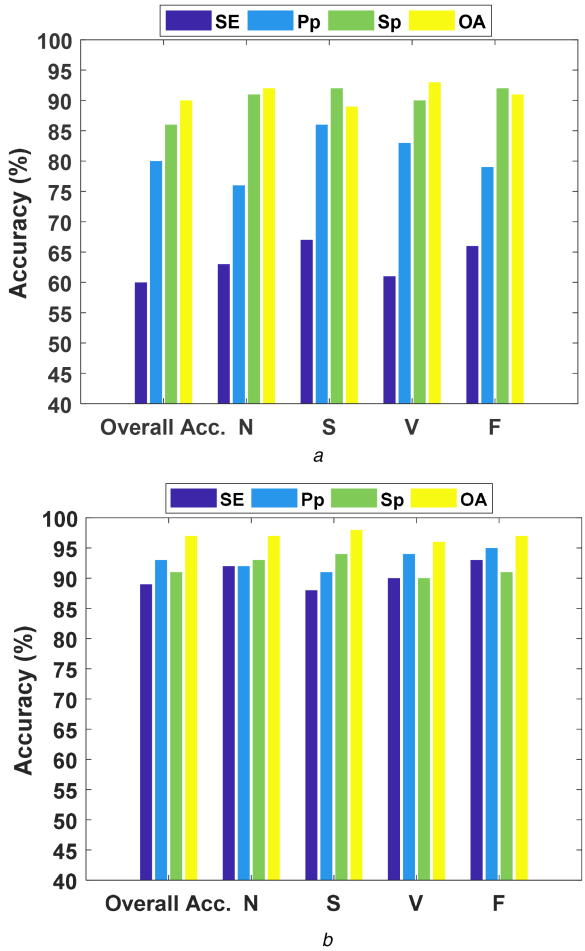

Fig. 8 *Comparison of standard NN and the proposed eigen-based DNN for N, S, V and F heartbeats classes*
(a) Standard feed-forward NN, ***(b)*** Proposed eigen-based DNN

Table 6 Learning parameters of classifiers for used databases

Classifiers	Learning parameters
NB	debug: True, displayModelInOldFormat: True, numDecimalPlaces:10, batchSize: 10000
SVM	sigma:0.5, Kernel:Puk, Epsilon: 1.0E-10, cacheSize: 100,000, omega:0.5
DT	numDecimalPlaces: 100, batchSize: 200, doNotCheckCapabilities: False
RF	numIterations: 200, numDecimalPlaces: 2, numExecutionSlots: 5, seed: 22

practical applications, classification procedure should be performed on datasets that are formed from completely different heartbeats.

4.4 Experiment III: comparison with the other classifiers

In this paper, evaluation dataset 3 which had 44 records (i.e. DS1 + DS2) is classified by using four different classifiers. It should be emphasised that the only the first four layers of the DNN are used to first calculate the ECG feature vectors and then four conventional classifiers are used instead of the softmax regression layer. These classifiers commonly used in the literature are NB, linear SVM, RF and DT. Also, the parameters of all classifiers are given in Table 6.

The classification performances of these classifiers and DNN-Entropy are presented in Table 6. According to the results shown in Table 7, the best ACC value is obtained by DNN-entropy model. Specifically, by using the SVM, the OA value is obtained as 91 and 93.2%, which are far greater than the OA values that are obtained by the end-to-end DNN-entropy architecture. It should be noted

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Table 7 Classification results in terms of VEB and SVEB Using D1 + DS2 of MIT-BIH

Classifiers	SVEB				VEB			
	Se	Pp	Sp	OA	Se	Pp	Sp	OA
NB	90.3	92.4	96.5	95.7	93.3	88.4	91.6	96.4
SVM	88.2	94.3	90.1	91.0	92.4	96.7	93.8	93.2
DT	85.2	87.3	94.4	90.3	91.4	89.3	86.4	89.4
RF	82.9	91.2	90.7	89.3	86.5	84.6	90.4	87.3
Eig-DNN-entropy	96.3	98.4	100.0	99.8	99.5	99.8	98.6	99.5

that linear SVM cannot produce effective results as the Gaussian SVM. However, the ACC value of linear SVM is higher than that is generated by the RF, NB and DT classifiers.

5 Conclusion

This paper proposes a novel DNN architecture for ECG signal classification by applying the AL and the eigenvalues of ECG beats. The features which are calculated from raw ECG signals are used as input to DAE. Both the magnitude of the eigenvalues and statistical-morphological features such as RR-interval are fused. Hybrid features are fed to DAE for building a sparse feature representation. In the feature learning phase, DAE is used to calculate the underlying features of ECG beats. A softmax classifier is used for efficient ECG classification, which is trained with ECG records of MIT-BIH data set.

The proposed method has some advantages: (ii) the most relevant ECG signals are selected with AL methodology; (ii) the proposed DNN model is capable of operating with eigenvalues and statistical features to obtain sparse feature representation via DAE; and (iii) our method can provide a patient-specific treatment model since it contains complete information on ECG beat.

In accordance with AAMI standards, ECG signals have been classified into four different classes. Classification performance is presented using four common measurements (sensitivity, specificity, positive predictivity and overall accuracy). Evaluation results on MIT-BIH dataset confirm that the proposed method achieves numerically more feasible ECG classification performance.

There are also some limitations to the study. Compared with existing models, feature learning phase of the proposed model has more time consuming procedure. Another limitation is the effect of temporal waveform patterns on the feature extraction. In the following work, the ECG classification performance of this model will be further improved in terms of various factors such as temporal patterns variations and computational complexity.

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